

Sato, Takashi

professor

基本信息

姓名: Sato, Takashi

所属院校: 京都大学

地区: Japan

学科领域

电子与计算机工程

集成电路设计与集成系统

微电子科学与工程

大功率半导体科学与工程

信息安全

人工智能

研究领域

EDA工具与电路验证

电路与系统设计

计算机体系结构

低功耗电路与能量管理

数字集成电路设计

功率器件物理与结构

微电子器件与制造

可靠性测试与失效分析

深度学习与神经网络

密码学与密码分析

+6

个人简介

Takashi Sato is a Professor at the Department of Informatics, Graduate School of Informatics. He holds a Master of Engineering from Waseda University and a Doctorate in Informatics from Kyoto University. His research focuses on the design and optimization of integrated circuits, reliability-aware and variability-aware circuit design, hardware security, power device modeling, and emerging computing technologies. Sato has published extensively in top-tier journals and conferences, contributing to advancements in VLSI design, machine learning for CAD, and secure hardware systems. His work is accessible via multiple academic websites and research platforms.

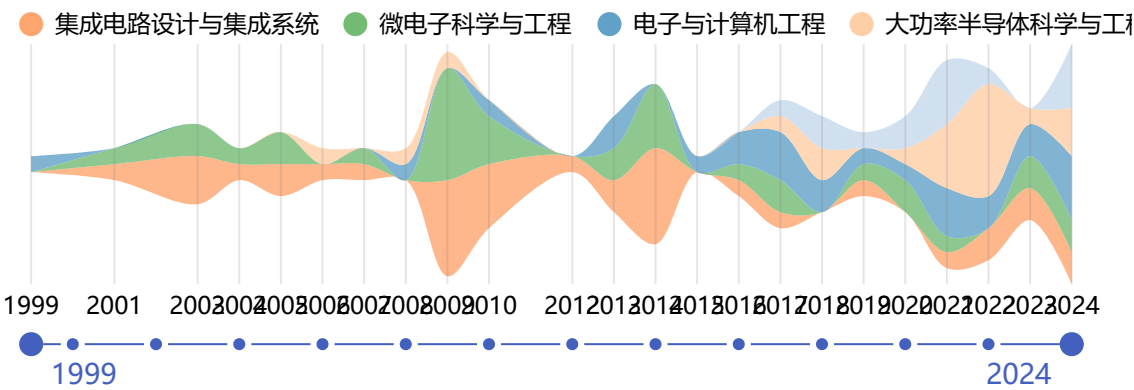
工作经历

京都大学 / Kyoto University - 教授 / Professor	2009-04-01 -
Kyoto University	2009-04-01 -
Tokyo Institute of Technology	2006-03-01 - 2009-03-31

教育经历

早稻田大学 - 工学修士	null - null
京都大学 - 博士 (情報学)	null - null

近年发表的论文



Low voltage operation of organic thin-film transistor with atmospheric coating of high-k polymer dielectric
JAPANESE JOURNAL OF APPLIED PHYSICS
Kuribara, Kazunori; Takei, Atsushi; Sato, Takashi; Yoshida, Manabu
被引用次数: 1 2024

BayesianSpikeFusion: accelerating spiking neural network inference via Bayesian fusion of early prediction
FRONTIERS IN NEUROSCIENCE
Habara, Takehiro; Sato, Takashi; Awano, Hiromitsu
被引用次数: 0 2024

(Invited) Redefining Outliers for On-Wafer Electrical Testing
PROCEEDINGS OF THE 2024 ACM/IEEE INTERNATIONAL SYMPOSIUM ON MACHINE LEARNING FOR CAD, MLCAD 2024
Shintani, Michihiro; Sato, Takashi
被引用次数: 0 2024

(Invited) Redefining Outliers for On-Wafer Electrical Testing

2024 ACM/IEEE 6TH SYMPOSIUM ON MACHINE LEARNING FOR CAD, MLCAD 2024

Shintani, Michihiro; Sato, Takashi

被引用次数: 0

2024

(Invited) Redefining Outliers for On-Wafer Electrical Testing

2024 ACM/IEEE 6TH SYMPOSIUM ON MACHINE LEARNING FOR CAD, MLCAD 2024

Shintani, Michihiro; Sato, Takashi

被引用次数: 0

2024

Fast Parameter Optimization of Delayed Feedback Reservoir with Backpropagation and Gradient Descent

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Ikeda, Sosei; Awano, Hiromitsu; Sato, Takashi

被引用次数: 0

2024

DNA-based Similar Image Retrieval via Triplet Network-driven Encoder

2024 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Koike, Takefumi; Awano, Hiromitsu; Sato, Takashi

被引用次数: 0

2024

CLAPPER: Clonable LFSR-based Asymmetric PUF-group with Peer-to-peer Equivalent Response

2024 IEEE 67TH INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS, MWSCAS 2024

Chen, Zhenzhe; Liu, Kunyang; Shinohara, Hirofumi; Sato, Takashi

被引用次数: 0

2024

Double MAC on a Cell: A 22-nm 8T-SRAM-Based Analog In-Memory Accelerator for Binary/Ternary Neural Networks Featuring Split Wordline

IEEE OPEN JOURNAL OF CIRCUITS AND SYSTEMS

Tagata, Hiroto; Sato, Takashi; Awano, Hiromitsu

被引用次数: 0

2024

SpongePUF: A Modeling Attack Resilient Strong PUF with Scalable Challenge Response Pair

2024 IEEE INTERNATIONAL SYMPOSIUM ON HARDWARE ORIENTED SECURITY AND TRUST, HOST

Chen, Zhenzhe; Sato, Takashi; Shinohara, Hirofumi

被引用次数: 0

2024

HOGGE : Homomorphic Gate on An FPGA

29TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2024

Matsuoka, Kotaro; Bian, Song; Sato, Takashi

被引用次数: 0

2024

Design of Aging-Robust Clonable PUF Using an Insulator-Based ReRAM for Organic Circuits

29TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2024

Oshima, Kunihiro; Kuribara, Kazunori; Sato, Takashi

被引用次数: 0

2024

Gaussian process-based device model toward a unified current model across room to cryogenic temperatures

2024 IEEE 36TH INTERNATIONAL CONFERENCE ON MICROELECTRONIC TEST STRUCTURES, ICMTS 2024

Shintani, Michihiro; Iwasaki, Tetsuro; Sato, Takashi

被引用次数: 0

2024

Logic Locking over TFHE for Securing User Data and Algorithms

29TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2024

Suemitsu, Kohei; Matsuoka, Kotaro; Sato, Takashi; Hashimoto, Masanori

被引用次数: 0

2024

Triplet Network-Based DNA Encoding for Enhanced Similarity Image Retrieval

Design Automation Conference

Takefumi Koike; Hiromitsu Awano; Takashi Sato

被引用次数: 0

2024

Aging-robust amplifier composed of p-type low voltage OTFT and organic semiconductor load

JAPANESE JOURNAL OF APPLIED PHYSICS

Kaneiwa, Yuto; Kuribara, Kazunori; Sato, Takashi

被引用次数: 1

2023

Measurement of 64 organic thin-film transistors in an array test structure using a relay-switch board for efficient evaluation of long-term reliability

JAPANESE JOURNAL OF APPLIED PHYSICS

Ogasahara, Yasuhiro; Kuribara, Kazunori; Sato, Takashi

被引用次数: 0

2023

Uncertainty-Aware Haptic Shared Control With Humanoid Robots for Flexible Object Manipulation

IEEE ROBOTICS AND AUTOMATION LETTERS

Hara, Takumi; Sato, Takashi; Ogata, Tetsuya; Awano, Hiromitsu

被引用次数: 2

2023

Modular DFR: Digital Delayed Feedback Reservoir Model for Enhancing Design Flexibility

ACM TRANSACTIONS ON EMBEDDED COMPUTING SYSTEMS

Ikeda, Sosei; Awano, Hiromitsu; Sato, Takashi

被引用次数: 2

2023

Dominant Model-Parameter Determination for the Analysis of Current Imbalance Across Paralleled Power Transistors

IEEE TRANSACTIONS ON POWER ELECTRONICS

Nakamura, Yohei; Shintani, Michihiro; Sato, Takashi

被引用次数: 0	2023
Hybrid CMOS and Pseudo-CMOS Organic Memory for Flexible Sensors	
IEEE SENSORS JOURNAL	
Qin, Zhaoxing; Kuribara, Kazunori; Ogasahara, Yasuhiro; Sato, Takashi	
被引用次数: 1	2023
Flex-SNN: Spiking Neural Network on Flexible Substrate	
IEEE SENSORS LETTERS	
Oshima, Kunihiro; Kuribara, Kazunori; Sato, Takashi	
被引用次数: 0	2023
Invited Paper: Overview of 2023 CAD Contest at ICCAD	
2023 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD	
Sato, Takashi; Wang, Chun-Yao; Chen, Yu-Guang; Huang, Tsung-Wei	
被引用次数: 0	2023
Efficient Analysis for Mitigation of Workload-Dependent Aging Degradation	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Morita, Shumpei; Bian, Song; Shintani, Michihiro; Sato, Takashi	
被引用次数: 1	2022
VisualNet: An End-to-End Human Visual System Inspired Framework to Reduce Inference Latency of Deep Neural Networks	
IEEE TRANSACTIONS ON COMPUTERS	
Wang, Tianchen; Zhang, Jiawei; Xiong, Jinjun; Bian, Song; Yan, Zheyu; Huang, Meiping; Zhuang, Jian; Sato, Takashi; Xu, Xiaowei; Shi, Yiyu	
被引用次数: 1	2022
Evaluation of thermal couple impedance model of power modules for accurate die temperature estimation up to 200 °C	
JAPANESE JOURNAL OF APPLIED PHYSICS	
Nakamura, Yohei; Kuroda, Naotaka; Nakahara, Ken; Shintani, Michihiro; Sato, Takashi	
被引用次数: 0	2022
Yield and leakage current of organic thin-film transistor logic gates toward reliable and low-power operation of large-scale logic circuits for IoT nodes	
JAPANESE JOURNAL OF APPLIED PHYSICS	
Ogasahara, Yasuhiro; Kuribara, Kazunori; Oshima, Kunihiro; Qin, Zhaoxing; Sato, Takashi	
被引用次数: 2	2022
Hardware-Friendly Delayed-Feedback Reservoir for Multivariate Time-Series Classification	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Ikeda, Sosei; Awano, Hiromitsu; Sato, Takashi	
被引用次数: 2	2022

Accelerating Parameter Extraction of Power MOSFET Models Using Automatic Differentiation

IEEE TRANSACTIONS ON POWER ELECTRONICS

Shintani, Michihiro; Ueda, Aoi; Sato, Takashi

被引用次数: 4

2022

Adaptive Outlier Detection for Power MOSFETs Based on Gaussian Process Regression

2022 IEEE APPLIED POWER ELECTRONICS CONFERENCE AND EXPOSITION, APEC

Shimozato, Kyohei; Shintani, Michihiro; Sato, Takashi

被引用次数: 3

2022

Layout-Dependent Vertical and In-Plane Leakage Current Reduction of Organic Thin-Film Transistors by Layer Contact Restriction

2022 IEEE 34TH INTERNATIONAL CONFERENCE ON MICROELECTRONIC TEST STRUCTURES (ICMTS)

Oshima, Kunihiro; Kuribara, Kazunori; Sato, Takashi

被引用次数: 1

2022

SASIMI: Evaluation Board for EM Information Leakage from Large Scale Cryptographic Circuits

2022 IEEE INTERNATIONAL SYMPOSIUM ON ELECTROMAGNETIC COMPATIBILITY & SIGNAL/POWER INTEGRITY, EMCSI

Fujimoto, Daisuke; Kim, Youngwoo; Hayashi, Yuichi; Homma, Naofumi; Hashimoto, Masanori; Sato, Takashi; Danger, Jean-Luc

被引用次数: 1

2022

Characteristic Degradation of Power MOSFETs by X-Ray Irradiation and their Recovery

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Shiozaki, Masato; Sato, Takashi

被引用次数: 3

2022

Characteristic Degradation of Power MOSFETs by X-Ray Irradiation and Their Recovery

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Shiozaki, Masato; Sato, Takashi

被引用次数: 0

2022

Gate Input Capacitance Characterization for Power MOSFETs Using Turn-on and Turn-off Switching Waveforms

2022 24TH EUROPEAN CONFERENCE ON POWER ELECTRONICS AND APPLICATIONS (EPE'22 ECCE EUROPE)

Nishitani, Yota; Inoue, Michiko; Sato, Takashi; Shintani, Michihiro

被引用次数: 0

2022

APAS: Application-Specific Accelerators for RLWE-Based Homomorphic Linear Transformations

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Bian, Song; Kundi, Dur E. Shahwar; Hirozawa, Kazuma; Liu, Weiqiang; Sato, Takashi

被引用次数: 6	2021
Separation of bias stress degradation between insulator and semiconductor carrier trapping in organic thin-film transistors	
JAPANESE JOURNAL OF APPLIED PHYSICS	
Oshima, Kunihiro; Bian, Song; Kuribara, Kazunori; Sato, Takashi	
被引用次数: 2	2021
An electrothermal compact model of SiC MOSFETs for analyzing avalanche failure mechanisms	
JAPANESE JOURNAL OF APPLIED PHYSICS	
Shimozato, Kyohei; Nakamura, Yohei; Bian, Song; Sato, Takashi	
被引用次数: 1	2021
Stable organic SRAM cell with p-type access transistors	
JAPANESE JOURNAL OF APPLIED PHYSICS	
Qin, Zhaoxing; Bian, Song; Kuribara, Kazunori; Sato, Takashi	
被引用次数: 3	2021
Virtual Secure Platform: A Five-Stage Pipeline Processor over TFHE	
USENIX Security Symposium	
Kotaro Matsuoka; Ryotaro Banno; Naoki Matsumoto; Takashi Sato; Song Bian	
被引用次数: 0	2021
Privacy-Preserving Medical Image Segmentation via Hybrid Trusted Execution Environment	
Design Automation Conference	
Song Bian; Weiwen Jiang; Takashi Sato	
被引用次数: 0	2021
Motion Robust Remote Photoplethysmography via Frequency Domain Motion Artifact Reduction	
2021 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE (IEEE BIOCAS 2021)	
Hebbar, Suraj; Sato, Takashi	
被引用次数: 2	2021
Dominant Model Parameter Extraction for Analyzing Current Imbalance in Parallel Connected SiC MOSFETs	
2021 IEEE ENERGY CONVERSION CONGRESS AND EXPOSITION (ECCE)	
Nakamura, Yohei; Shintani, Michihiro; Sato, Takashi	
被引用次数: 5	2021
Statistical Device Modeling with Arbitrary Model-Parameter Distribution via Markov Chain Monte Carlo	
2021 INTERNATIONAL CONFERENCE ON SIMULATION OF SEMICONDUCTOR PROCESSES AND DEVICES (SISPAD 2021)	
Tsukamoto, Hiroki; Bian, Song; Sato, Takashi	

被引用次数: 0

2021

Analysis of Thermal Concentration Failure in Unclamped Inductive Switching Based on Three-Dimensional Electro-Thermal Simulation With On-Chip Variation

2021 INTERNATIONAL CONFERENCE ON SIMULATION OF SEMICONDUCTOR PROCESSES AND DEVICES (SISPAD 2021)

Shimozato, Kyohei; Nakamura, Yohei; Sato, Takashi

被引用次数: 2

2021

Privacy-Preserving Medical Image Segmentation via Hybrid Trusted Execution Environment

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Bian, Song; Jiang, Weiwen; Sato, Takashi

被引用次数: 4

2021

Overview of 2021 CAD Contest at ICCAD

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Huang, Tsung-Wei; Chen, Yu-Guang; Wang, Chun-Yao; Sato, Takashi

被引用次数: 0

2021

Virtual Secure Platform: A Five-Stage Pipeline Processor over TFHE

PROCEEDINGS OF THE 30TH USENIX SECURITY SYMPOSIUM

Matsuoka, Kotaro; Banno, Ryotaro; Matsumoto, Naoki; Sato, Takashi; Bian, Song

被引用次数: 16

2021

Ed-PUF: Event-Driven Physical Unclonable Function for Camera Authentication in Reactive Monitoring System

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zheng, Yue; Zhao, Xiaojin; Sato, Takashi; Cao, Yuan; Chang, Chip-Hong

被引用次数: 10

2020

Recovery-aware bias-stress degradation model for organic thin-film transistors considering drain and gate bias voltages

JAPANESE JOURNAL OF APPLIED PHYSICS

Oshima, Kunihiro; Shintani, Michihiro; Kuribara, Kazunori; Ogasahara, Yasuhiro; Sato, Takashi

被引用次数: 7

2020

Statistical Extraction of Normally and Lognormally Distributed Model Parameters for Power MOSFETs

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Tsukamoto, Hiroki; Shintani, Michihiro; Sato, Takashi

被引用次数: 4

2020

Measurement and Modeling of Ambient-Air-Induced Degradation in Organic Thin-Film Transistor

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Shintani, Michihiro; Saito, Michiaki; Kuribara, Kazunori; Ogasahara, Yasuhiro; Sato, Takashi

被引用次数: 5	2020
Organic Current Mirror PUF for Improved Stability Against Device Aging	
IEEE SENSORS JOURNAL	
Qin, Zhaoxing; Shintani, Michihiro; Kuribara, Kazunori; Ogasahara, Yasuhiro; Sato, Takashi	
被引用次数: 11	2020
Clustering Approach for Solving Traveling Salesman Problems via Ising Model Based Solver	
Design Automation Conference	
Akira Dan; Riu Shimizu; Takeshi Nishikawa; Song Bian; Takashi Sato	
被引用次数: 0	2020
Feasibility of a low-power, low-voltage complementary organic thin film transistor buskeeper physical unclonable function	
JAPANESE JOURNAL OF APPLIED PHYSICS	
Ogasahara, Yasuhiro; Kuribara, Kazunori; Shintani, Michihiro; Sato, Takashi	
被引用次数: 9	2019
Hardware-Accelerated Secured Naive Bayesian Filter Based on Partially Homomorphic Encryption	
IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES	
Bian, Song; Hiromoto, Masayuki; Sato, Takashi	
被引用次数: 0	2019
GPU-based Ising computing for solving max-cut combinatorial optimization problems	
INTEGRATION-THE VLSI JOURNAL	
Cook, Chase; Zhao, Hengyang; Sato, Takashi; Hiromoto, Masayuki; Tan, Sheldon X. -D.	
被引用次数: 20	2019
Mechanically and electrically robust metal-mask design for organic CMOS circuits	
JAPANESE JOURNAL OF APPLIED PHYSICS	
Shintani, Michihiro; Qin, Zhaoxing; Kuribara, Kazunori; Ogasahara, Yasuhiro; Hiromoto, Masayuki; Sato, Takashi	
被引用次数: 6	2018
Area Efficient Annealing Processor for Ising Model without Random Number Generator	
IEICE TRANSACTIONS ON INFORMATION AND SYSTEMS	
Gyoten, Hidenori; Hiromoto, Masayuki; Sato, Takashi	
被引用次数: 23	2018
MRO-PUF: Physically Unclonable Function with Enhanced Resistance against Machine Learning Attacks Utilizing Instantaneous Output of Ring Oscillator	
IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES	
Hiromoto, Masayuki; Yoshinaga, Motoki; Sato, Takashi	

被引用次数: 1

2018

Efficient Mini-Batch Training on Memristor Neural Network Integrating Gradient Calculation and Weight Update

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Yamamori, Satoshi; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 1

2018

Surface-Potential-Based Silicon Carbide Power MOSFET Model for Circuit Simulation

IEEE TRANSACTIONS ON POWER ELECTRONICS

Shintani, Michihiro; Nakamura, Yohei; Oishi, Kazuki; Hiromoto, Masayuki; Hikihara, Takashi; Sato, Takashi

被引用次数: 23

2018

Coin Flipping PUF: A Novel PUF With Improved Resistance Against Machine Learning Attacks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Tanaka, Yuki; Bian, Song; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 26

2018

PARHELIA: Particle Filter-Based Heart Rate Estimation From Photoplethysmographic Signals During Physical Exercise

IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING

Fujita, Yuya; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 40

2018

Measurement and modeling of gate-drain capacitance of silicon carbide vertical double-diffused MOSFET

JAPANESE JOURNAL OF APPLIED PHYSICS

Shintani, Michihiro; Nakamura, Yohei; Hiromoto, Masayuki; Hikihara, Takashi; Sato, Takashi

被引用次数: 13

2017

Identification and Application of Invariant Critical Paths under NBTI Degradation

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Bian, Song; Morita, Shumpei; Shintani, Michihiro; Awano, Hiromitsu; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 0

2017

Efficient Aging-Aware Failure Probability Estimation Using Augmented Reliability and Subset Simulation

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Awano, Hiromitsu; Sato, Takashi

被引用次数: 0

2017

Utilization of Path-Clustering in Efficient Stress-Control Gate Replacement for NBTI Mitigation

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Morita, Shumpei; Bian, Song; Shintani, Michihiro; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 0

2017

RTN in Scaled Transistors for On-Chip Random Seed Generation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Mohanty, Abinash; Sutaria, Ketul B.; Awano, Hiromitsu; Sato, Takashi; Cao, Yu

被引用次数: 22

2017

Scalable Device Array for Statistical Characterization of BTI-Related Parameters

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Awano, Hiromitsu; Morita, Shumpei; Sato, Takashi

被引用次数: 3

2017

Efficient Aging-Aware SRAM Failure Probability Calculation via Particle Filter-Based Importance Sampling

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Awano, Hiromitsu; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 2

2016

Fast Estimation of NBTI-Induced Delay Degradation Based on Signal Probability

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Bian, Song; Shintani, Michihiro; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 2

2016

Path Clustering for Test Pattern Reduction of Variation-Aware Adaptive Path Delay Testing

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Shintani, Michihiro; Uezono, Takumi; Hatayama, Kazumi; Masu, Kazuya; Sato, Takashi

被引用次数: 1

2016

Stabilization of a high-order harmonic generation seeded extreme ultraviolet free electron laser by time-synchronization control with electro-optic sampling

HIGH POWER LASER SCIENCE AND ENGINEERING

Tomizawa, H.; Sato, T.; Ogawa, K.; Togawa, K.; Tanaka, T.; Hara, T.; Yabashi, M.; Tanaka, H.; Ishikawa, T.; Togashi, T.; Matsubara, S.; Okayasu, Y.; Watanabe, T.; Takahashi, E. J.; Midorikawa, K.; Aoyama, M.; Yamakawa, K.; Owada, S.; Iwasaki, A.; Yamanouchi, K.

被引用次数: 8

2015

Introduction to: Special Issue on Cross-Layer System Design

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Shi, Yiyu; Sato, Takashi

被引用次数: 0

2015

An Error Correction Scheme through Time Redundancy for Enhancing Persistent Soft-Error Tolerance of CGRAs

IEICE TRANSACTIONS ON ELECTRONICS

Imagawa, Takashi; Hiromoto, Masayuki; Ochi, Hiroyuki; Sato, Takashi

被引用次数: 0

2015

A Variability-Aware Adaptive Test Flow for Test Quality Improvement

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Shintani, Michihiro; Uezono, Takumi; Takahashi, Tomoyuki; Hatayama, Kazumi; Aikyo, Takashi; Masu, Kazuya; Sato, Takashi

被引用次数: 18

2014

Automation of Model Parameter Estimation for Random Telegraph Noise

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Shimizu, Hirofumi; Awano, Hiromitsu; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 2

2014

IDDQ Outlier Screening through Two-Phase Approach: Clustering-Based Filtering and Estimation-Based Current-Threshold Determination

IEICE TRANSACTIONS ON INFORMATION AND SYSTEMS

Shintani, Michihiro; Sato, Takashi

被引用次数: 0

2014

BTlarray: A Time-Overlapping Transistor Array for Efficient Statistical Characterization of Bias Temperature Instability

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Awano, Hiromitsu; Hiromoto, Masayuki; Sato, Takashi

被引用次数: 10

2014

Aging Statistics Based on Trapping/Detrapping: Compact Modeling and Silicon Validation

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Sutaria, Ketul B.; Velamala, Jyothi Bhaskarr; Kim, Chris H.; Sato, Takashi; Cao, Yu

被引用次数: 23

2014

Hypersphere Sampling for Accelerating High-Dimension and Low-Failure Probability Circuit-Yield Analysis

IEICE TRANSACTIONS ON ELECTRONICS

Hagiwara, Shiho; Date, Takanori; Masu, Kazuya; Sato, Takashi

被引用次数: 1

2014

State-Dependence of On-Chip Power Distribution Network Capacitance

IEICE TRANSACTIONS ON ELECTRONICS

Yamanaga, Koh; Hagiwara, Shiho; Takahashi, Ryo; Masu, Kazuya; Sato, Takashi

被引用次数: 1

2014

Device-Parameter Estimation through IDDQ Signatures

IEICE TRANSACTIONS ON INFORMATION AND SYSTEMS

Shintani, Michihiro; Sato, Takashi

被引用次数: 1

2013

Compact Modeling of Statistical BTI under Trapping/Detrapping

IEEE TRANSACTIONS ON ELECTRON DEVICES

Velamala, Jyothi Bhaskarr; Sutaria, Ketul B.; Shimizu, Hirofumi; Awano, Hiromitsu; Sato, Takashi; Wirth, Gilson; Cao, Yu

被引用次数: 63

2013

Static properties of planar Mach-Zehnder optical modulator on (110) InP substrate

ELECTRONICS LETTERS

Ogiso, Y.; Arai, M.; Sato, T.; Shibata, Y.; Kohtoku, M.

被引用次数: 2

2013

A Cost-Effective Selective TMR for Coarse-Grained Reconfigurable Architectures Based on DFG-Level Vulnerability Analysis

IEICE TRANSACTIONS ON ELECTRONICS

Imagawa, Takashi; Tsutsui, Hiroshi; Ochi, Hiroyuki; Sato, Takashi

被引用次数: 2

2013

Parallel Acceleration Scheme for Monte Carlo Based SSTA Using Generalized STA Processing Element

IEICE TRANSACTIONS ON ELECTRONICS

Yuasa, Hiroshi; Tsutsui, Hiroshi; Ochi, Hiroyuki; Sato, Takashi

被引用次数: 2

2013

A Variability-Aware Energy-Minimization Strategy for Subthreshold Circuits

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Kawashima, Junya; Tsutsui, Hiroshi; Ochi, Hiroyuki; Sato, Takashi

被引用次数: 1

2012

Power Distribution Network Optimization for Timing Improvement with Statistical Noise Model and Timing Analysis

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Enami, Takashi; Sato, Takashi; Hashimoto, Masanori

被引用次数: 0

2012

Bayesian Estimation of Multi-Trap RTN Parameters Using Markov Chain Monte Carlo Method

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

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